

Predicting Post-Earnings Announcement Drift Using NLP

Daniel Wang

w266: Natural Language Processing with Deep Learning
August 6, 2025

Objective

To determine whether you can predict the direction of the stock price movement after an earnings call using the transcript text.

Full Conference Call Transcript

Lip Bu will open with comments on our second quarter results, as well as provide an update on our strategy and priorities. Dave will then discuss our overall financial results, including third quarter guidance, before we transition to answer your questions. Before we begin, please note that today's discussion contains forward-looking statements based on the environment as we currently see it and, as such, are subject to various risks and uncertainties. It also contains references to non-GAAP financial measures that we believe provide useful information to our investors.

Our earnings release, most recent annual report on Form 10-K, and other filings with the SEC provide more information on specific risk factors that could cause actual results to differ materially from our expectations. They also provide additional information on our non-GAAP financial measures, including reconciliations where appropriate to our corresponding GAAP financial measures. With that, let me turn things over to Lip Bu.

Lip Bu Tan: Thank you, John. And let me add my welcome. We had a solid Q2 with revenue above the high end of our guidance. This reflects strong demand across our business and good execution by the team. As expected, headline profitability was impacted by several one-time items and impairments. But I am pleased by the underlying operating performance in the quarter, even as we have more work to do. Dave will go through our detailed financials shortly. Today, I want to provide you with updates on four major initiatives where we have started to make progress and will continue to focus in coming quarters: our organization and culture, our

milestones and not before. As part of this new financial discipline, we have decided not to continue with our manufacturing projects in Germany and Poland. We also plan to consolidate our assembly and test operations in Costa Rica into larger existing sites in Vietnam and Malaysia. And we will further slow the pace of construction in Ohio to ensure our spending is aligned with market demand.

when I'm confident those returns exist. I'm intimately familiar with the foundry and fabulous ecosystem, having helped create it over the last two decades. I'm using that experience to put our Intel foundry on a more solid footing for the future.

Lip Bu Tan: Ross, thank you so much for a good question. First of all, we focus on the 18A. We, you know, so far the last two months, and I have twice a week review with the team. And you know we make steady

COST Shows Upward Drift After Negative Jump

Market Summary > Costco Wholesale Corp

955.37 USD

-87.51 (-8.39%) ↓ past 6 months

Closed: Aug 4, 6:42 PM EDT • Disclaimer

After hours 955.38 +0.010 (0.0010%)



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Transcript and stock data

	Date	Next_biz_day	Company_name	Ticker	Word_count	Text	Open	Close	Close_1	Close_2	Close_5	Close_0_dir	Close_1_dir	Close_2_dir	Close_5_dir
0	2015-01-06 21:30:00	2015-01-07	Micron Technology, Inc.	MU	5184	Good afternoon. My name is Karen, and I'll be ...	30.795428	31.352146	32.885567	32.748825	29.349901	1.0	1.0	1.0	0.0
1	2015-01-08 15:30:00	2015-01-09	Constellation Brands, Inc.	STZ	3188	Ladies and gentlemen, thank you for standing b...	92.797200	93.153008	92.831917	92.250504	96.849953	1.0	1.0	0.0	1.0
2	2015-01-12 22:00:00	2015-01-13	Alcoa Inc.	AA	760	Good day, ladies and gentlemen, and welcome to...	37.120452	35.610390	33.694641	33.739716	35.880848	0.0	0.0	0.0	0.0
3	2015-01-14 13:30:00	2015-01-15	CSX Corporation	CSX	1909	Good morning, ladies and gentlemen, and welcom...	9.566863	9.651751	9.863967	9.773422	9.753616	1.0	1.0	1.0	1.0
4	2015-01-14 13:30:00	2015-01-15	JPMorgan Chase & Co.	JPM	2758	Good morning, ladies and gentlemen, and welcom...	41.869368	41.313416	42.019623	41.854340	42.583084	0.0	1.0	0.0	1.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
17228	2024-12-19 16:00:00	2024-12-20	FactSet Research Systems Inc.	FDS	2585	Good day, and thank you for standing by. Welco...	482.339360	481.274323	487.067291	489.067993	480.438232	0.0	1.0	1.0	0.0

Binary classification task

1: Upward price movement

0: Downward price movement

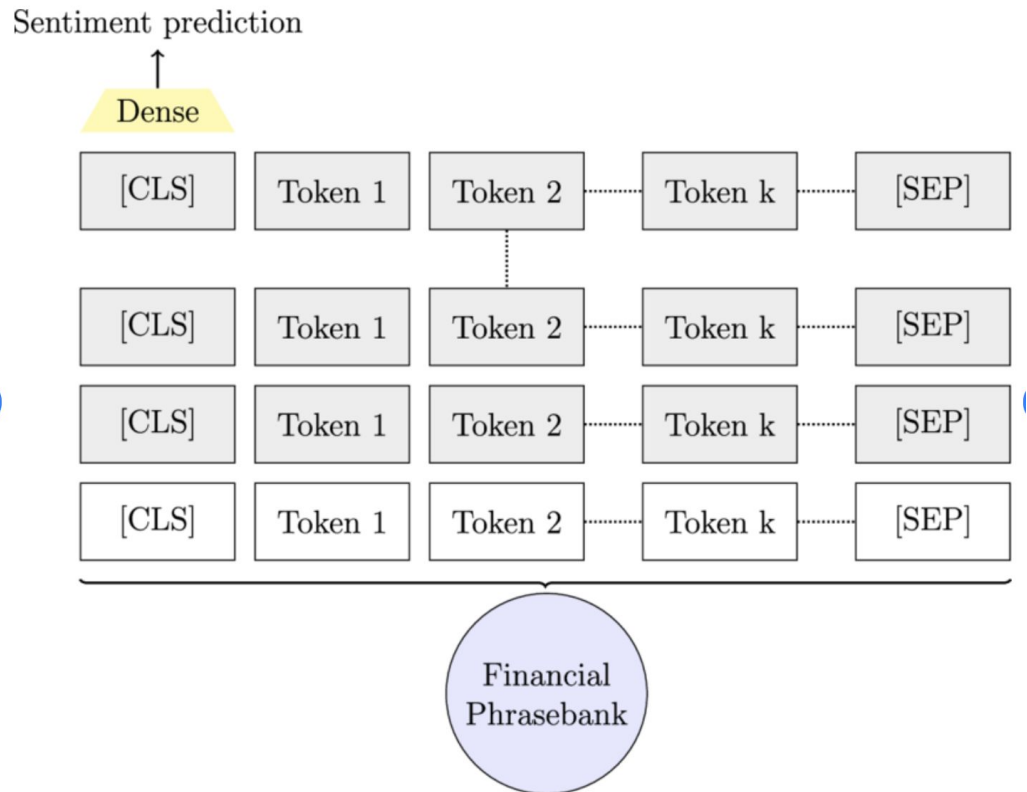
High majority baseline

1: Upward price movement	53.67%
0: Downward price movement	46.33%

Models experimented with

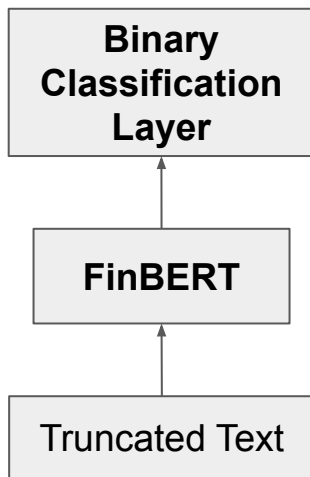
Model	Overall Accuracy	Up-Precision	Up-Recall	Up-F1
Majority baseline (Up always)	53.67%	N/A	N/A	N/A
ModernBERT				
FinBERT test 1 (classifier only)				
FinBERT test 2 (pooler + clf)				
FinBERT test 3 (above plus transformer layers 9-12)				
Chunked FinBERT + BiLSTM				
FinBERT - positive sentiment only subset				
LLaMA-based RF classifier				
Ensemble (FinBERT + LLaMA agree Up)				

FinBERT models

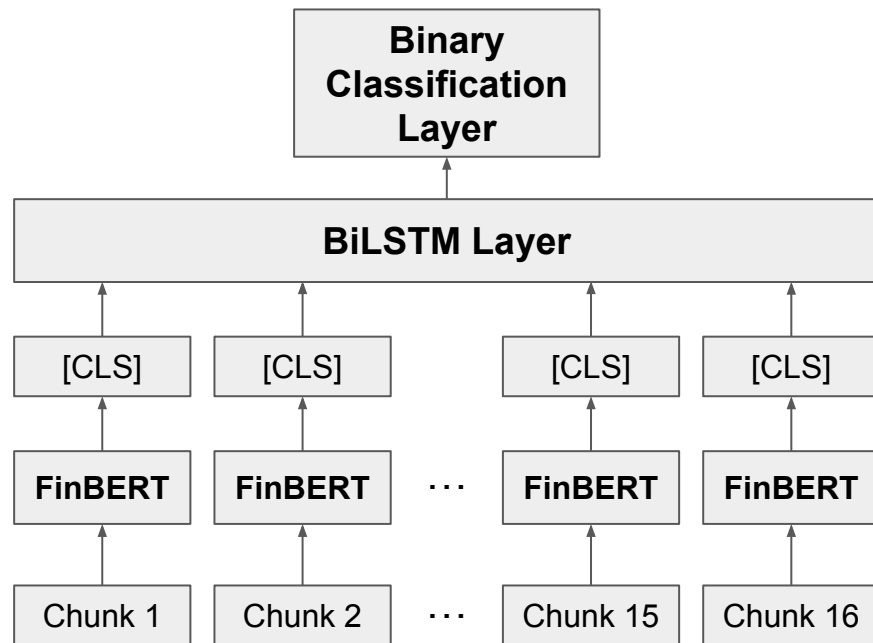


FinBERT models

Truncation



Chunking + BiLSTM



Model accuracies

Model	Avg Accuracy
Majority baseline	53.67%
FinBERT with truncated text	~52.8%
Chunked FinBERT + BiLSTM	~53.6%

Recall (Up) for chunked model at 99%

Model	Accuracy	Recall (Up)
Majority baseline	53.67%	--
FinBERT with truncated text	~52.8%	60%-90%
Chunked FinBERT + BiLSTM	~53.6%	~99%

Extremely few negative sentiment transcripts

Model	Accuracy	Recall (Up)
Majority baseline	53.67%	--
FinBERT with truncated text	~52.8%	60%-100%
Chunked FinBERT + BiLSTM	~53.6%	~99%
	Positive	Negative
% of Pos/Neg Sent. by FinBERT	>99%	<1%

LLaMA-3.1 with prompt engineering

```
4 # Getting the system and user messages
5 system_message = "You are a financial analyst assistant trained to interpret earnings call transcripts and predict near
6 user_message = f""Based on the following earnings call transcript, what is the likely one-week market direction of the
7 You are only allowed to respond with 2 words, which can be one of the following: confident up, confident down, unconfid
8 - confident up means that the underlying tone is still positive, and the speaker is showing little signs of hesitation
9 - unconfident up means that most of the transcript is confident, but perhaps with some minor points that could be hinti
10 - unconfident down means that the earnings call may project confidence, but the tone implies otherwise, and there may b
11 - confident down covers everything else, where even if the transcript is positive overall, if there's more hesitation,
12
13 Transcript:
14 {text_1}""
15
16 messages = [
17     {"role": "system", "content": system_message},
18     {"role": "user", "content": user_message}
19 ]
20
21 # Getting the outputs through a pipeline
22 outputs_1 = pipeline(messages, max_new_tokens=5)
23
24 pprint(outputs_1[0]["generated_text"][-1], compact=True)
```

Setting `pad\_token\_id` to `eos\_token\_id`:128001 for open-end generation.
{'content': 'confident up', 'role': 'assistant'}

Forced LLaMA outputs

4 outcomes:

“Confident up”

“Unconfident up”

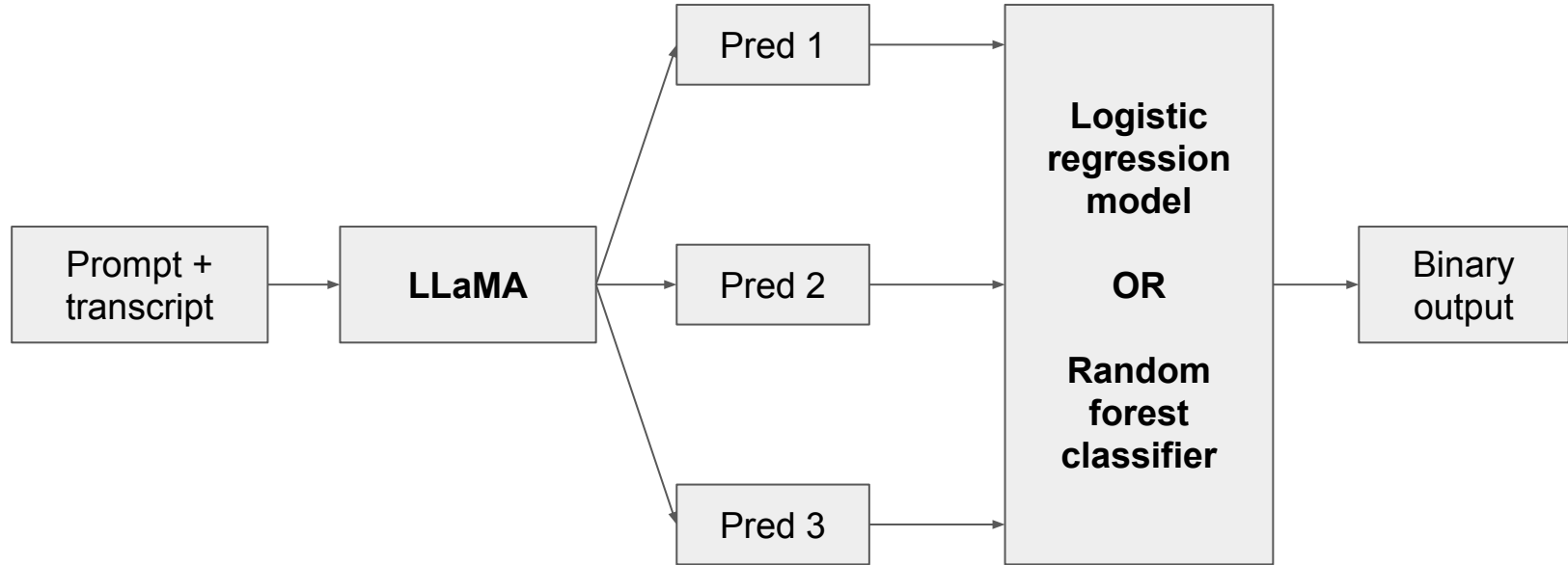
“Unconfident down”

“Confident down”

3 indep. predictions
per transcript

Transcript_Num		pred_1	pred_2	pred_3
0	0	unconfident down	unconfident up	unconfident up
1	1	confident up	unconfident up	confident up
2	2	confident up	confident up	unconfident down
3	3	confident up	confident up	confident up
4	4	unconfident down	confident up	confident up
...	...	...	...	...
17228	17228	unconfident down	confident up	confident up
17229	17229	unconfident down	unconfident down	unconfident down
17230	17230	unconfident up	unconfident down	confident down
17231	17231	confident up	unconfident up	confident down
17232	17232	unconfident up	confident up	confident up

LLaMA-based logistic regression / random forest classifier



LLaMA-based logistic/RF classifier accuracy

Model	Avg Accuracy
Majority baseline	53.67%
FinBERT with truncated text	~52.8%
Chunked FinBERT + BiLSTM	~53.6%
LLaMA-based logistic/RF classifier	~50.0%

Ensemble strategy with FinBERT and LLaMA predictions

% of Data

FinBERT Predictions	Up	Down
Up	46.4%	12.3%
Neutral / Down	29.2%	12.1%
LLaMA Predictions		
		Up Down

Strategy:

If both FinBERT and LLaMA agree on an “Up” prediction, then enter long into the trade.

Otherwise, abstain or default to majority class baseline.

Model accuracies

Model	Avg Accuracy
Majority baseline	53.67%
FinBERT with truncated text	~52.8%
Chunked FinBERT + BiLSTM	~53.6%
LLaMA-based logistic/RF classifier	~50.0%
Ensemble (FinBERT + LLaMA agree Up)	~54.6%

Conclusions

- Earnings calls are mostly “happy talk”, causing predictions to skew Up
- No singular model produced accuracy above baseline
 - Positive/negative sentiment does not necessarily translate into up/down stock price movement
- Ensemble strategy (long-only) between encoder and decoder models shows promise

Thank You

Appendix

Previous literature

An Exploratory Study of Stock Price Movements from Earnings Calls

Sourav Medya¹, Mohammad Rasoolinejad¹, Yang Yang², Brian Uzzi¹

¹Kellogg School of Management & Northwestern Institute on Complex Systems, Northwestern University

²Syracuse University

{sourav.medya, mohammad.rasoolinejad, uzzi}@kellogg.northwestern.edu
yyang87@sy.edu

ABSTRACT

Financial market analysis has focused primarily on extracting signals from accounting, stock price, and other numerical “hard” data reported in P&L statements or earnings per share reports. Yet, it is well-known that decision-makers routinely use “soft” text-based documents that interpret the hard data they narrate. Recent advances in computational methods for analyzing unstructured and soft text-based data at scale offer possibilities for understanding financial market behavior that could improve investments and market equity. A critical and ubiquitous form of soft data are earnings calls. Earnings calls are periodic (often quarterly) statements usually by CEOs who attempt to influence investors’ expectations of a company’s past and future performance. Here, we study the statistical relationship between earnings calls, company sales, stock performance, and analysts’ recommendations. Our study covers a decade of observations with approximately 100,000 transcripts of earnings calls from 6,300 public companies from January 2010 to December 2019. In this study, we report three novel findings. First, the buy, sell and hold recommendations from professional analysts made prior to the earnings call have low correlation with stock price movements after the earnings call. Second, using our graph neural network based method that processes the semantic features of earnings calls, we reliably and accurately predict stock price movements in five major areas of the economy. Third, the semantic features of transcripts are more predictive of stock price movements than sales and earnings per share, i.e., traditional hard data in most of the cases.

KEYWORDS

Earnings call, stock price movement, natural language processing

ACM Reference Format:

Sourav Medya¹, Mohammad Rasoolinejad¹, Yang Yang², Brian Uzzi¹. 2018. An Exploratory Study of Stock Price Movements from Earnings Calls. In *Woodstock ’18: ACM Symposium on Neural Gaze Detection*, June 03–05, 2018, Woodstock, NY, ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/1122445.1122456>

1 INTRODUCTION

Earnings calls are the discussion sessions that happen after the public companies release their financial data for the reporting period which is usually a quarter of the year or a fiscal year. Analysts, investors, and journalists are often present during the earnings calls and the recordings of these can be accessed through corresponding company websites.

In this session, usually the CEO and/or other representatives from the management of the company present their financial achievements during the last quarter and give guidelines for the next. The management usually discusses details about important company information such as growth, risks, purchases, liabilities, lawsuits, share buybacks, increase/decrease in dividends, any change in executive teams and future goals. The session usually consists of management discussion and analysis (MD&A section) followed by a question-answering session involving the audience and the investors.

The earnings call is a major event as the stock price market reflects higher level of volatility and trading volume prior, during and after earnings calls [9]. Such volatility can result in bad investments, missing profit opportunities and huge losses for the investors. The large amount of available data on transcripts and stock market prices give us an opportunity to predict the directions of stock price movements more accurately. Moreover, there are other factors such as sentiment on social media and news that can also affect the stock market and have been discussed in the literature [14, 36]. However, in this paper, we focus on associating the stock price movements from the transcripts of the earnings calls [28, 33].

Figure 1 shows an example of the stock price changes for the company Tesla, Inc. in 2019. The price movement after the occurrence of the earnings call can be positive or negative. As the figure shows daily effect, we define the label of the transcripts based on one day effect in the stock price (Def. 1 and Def. 2). While the price change is important, it is also meaningful to compare the change with respect to the broader market behavior. To achieve that we compare the rate of the stock price changes with the corresponding index (Table 1) over the next five business days. The labels of the transcripts are decided based on the majority of out-performances or under-performances with respect to the index value (Def. 3).



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Effect of Post-Earnings Announcement Drift with Textual and Contextual Factors of Earnings Calls

Andy Chung

Advanced Interdisciplinary Studies,
School of Engineering,
University of Tokyo
Tokyo, Japan
ly@ag.eccu-tokyo.ac.jp

Kumiko Tanaka-Ishii

Department of Computer Science and Engineering,
School of Fundamental Science and Engineering,
Waseda University
Tokyo, Japan
kumiko@waseda.jp

1 INTRODUCTION

The post-earnings announcement drift (PEAD), a well-known market anomaly [3], is the tendency for a stock price to drift toward an earnings surprise for a period after an earnings announcement. Inconsistent with the Efficient Market Hypothesis [13] states, the post-earnings announcement drift associated with earnings surprise does not lead to an immediate adjustment of the stock price but instead to a predictable drift of the stock price, which could last 60 trading days after the earnings announcement [16].

Previous studies have found that conference calls provide information to market participants in addition to the information in the corresponding press release and company filings [17]. However, previous studies related to post-earnings announcement drift modeling do not fully make use of earnings call information, for example, only to the extent of sentiment analysis of the entire corpus using the sentiment lexicon [37] [43] [44].

This paper proposes a systematic approach to derive textual and contextual features from earnings calls, including a text pre-processing procedure, ChatGPT for abstractive summarization, and a sentence-transformer-based contextual embedding extraction, followed by modeling PEAD with machine learning models. The main purpose of this paper is to examine the effectiveness of incorporating earnings calls into the PEAD prediction task.

Our primary contributions are the following:

- (1) We propose the earnings call factor, which consists of textual and contextual features derived from the earnings call, for PEAD prediction.
- (2) Our experiment shows that contextual features from earnings calls provide information unexplained by earnings, fundamental and technical features, improving models’ ability to predict PEAD.

2 RELATED WORK

2.1 Modeling post-earnings announcement drift

PEAD is one of the most robust and well-studied anomalies in the asset pricing literature. [22] shows evidence that transient institutional investors actively trade and profit from the PEAD anomaly. [23] propose adding a risk factor related to unexpected earnings surprise in Fama and French’s three-factor risk model [14]. They show

Post-Earnings Announcement Drift Prediction: Leveraging Investor Responses with Multitask Learning

Livia R. Liu Sheng*

Department of Management Information Systems, Lerner College of Business and Economics, University of Delaware,
Department of Information Systems, W. P. Carey School of Business, Arizona State University, Tempe,

<https://orcid.org/0000-0001-8278-9989> (YZ), slas.liu.108@asu.edu, <https://orcid.org/0000-0001-8277-4701> (XL);
<https://orcid.org/0000-0002-3763-2736> (CML)

2, 2024

Advance:

10558

Abstract. Post-earnings announcement drift (PEAD) refers to the phenomenon in which a company’s stock price tends to drift persistently in response to the information released during the earnings announcement event. Predicting PEAD is of great interest to both investors and researchers because the magnitude of PEAD is economically significant. Whereas decades of studies have explored various approaches to forecasting PEAD, prior research has largely overlooked postevent investor responses—a critical intermediary in the PEAD mechanism—because of the limitations of single-task learning (STL). This study addresses this gap by introducing a multitask learning (MTL) framework that explicitly incorporates investor responses as auxiliary tasks while mitigating look-ahead bias. To further enhance model performance, we propose GradPrep, an adaptive task weighting method that assigns greater weight to auxiliary tasks that contribute diverse and informative training signals. Our model employs a multilevel, multiquery transformer architecture to facilitate cross-task learning, effectively integrating structured financial features with lengthy earnings call transcripts. Evaluation of the model from 2010 to 2022 demonstrates that the proposed design not only outperforms benchmark models in terms of prediction accuracy but also generates a daily risk-adjusted return (alpha) two to three times larger than the traditional earnings surprise-based models. This study contributes to the stream of information systems (IS) literature at the intersection of artificial intelligence (AI), finance, and design science research. Our work provides valuable decision support modules and managerial implications for investment and other financial decision makers.

History: Eric Zheng, Senior Editor; Haimin Zhao, Associate Editor.

Supplemental Material: The online appendix is available at <https://doi.org/10.1287/isre.2022.0358>.

Post-earnings announcement drift (PEAD) • earnings calls transcripts • multitask learning (MTL) • adaptive weighting method • analytics • investment decision support

earnings announcement is one of the most widely used predictor of PEAD (Llunt and Mendenhall 2006). Trading strategies based on earnings surprises are reported to generate significant abnormal returns (ARs) (Bernard and Thomas 1989). However, because of its accessibility and extensive exploitation, recent literature suggests that the predictive power of earnings surprises on PEAD is diminishing (Chordia et al. 2009; Hung et al. 2015; Fink 2021; Martineau 2021).

A recent trend in PEAD prediction involves utilizing the unstructured content generated during earnings conference calls, commonly referred to as earnings calls, in addition to earnings surprises. A firm’s earnings call is typically conducted the same day or the day after the

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*AMP (actively managed funds) trade 170% more on earnings announcement (EA) days than on non-EA

INTC Shows Sideways Drift After Minor Negative Jump

Market Summary > Intel Corp

20.19 USD

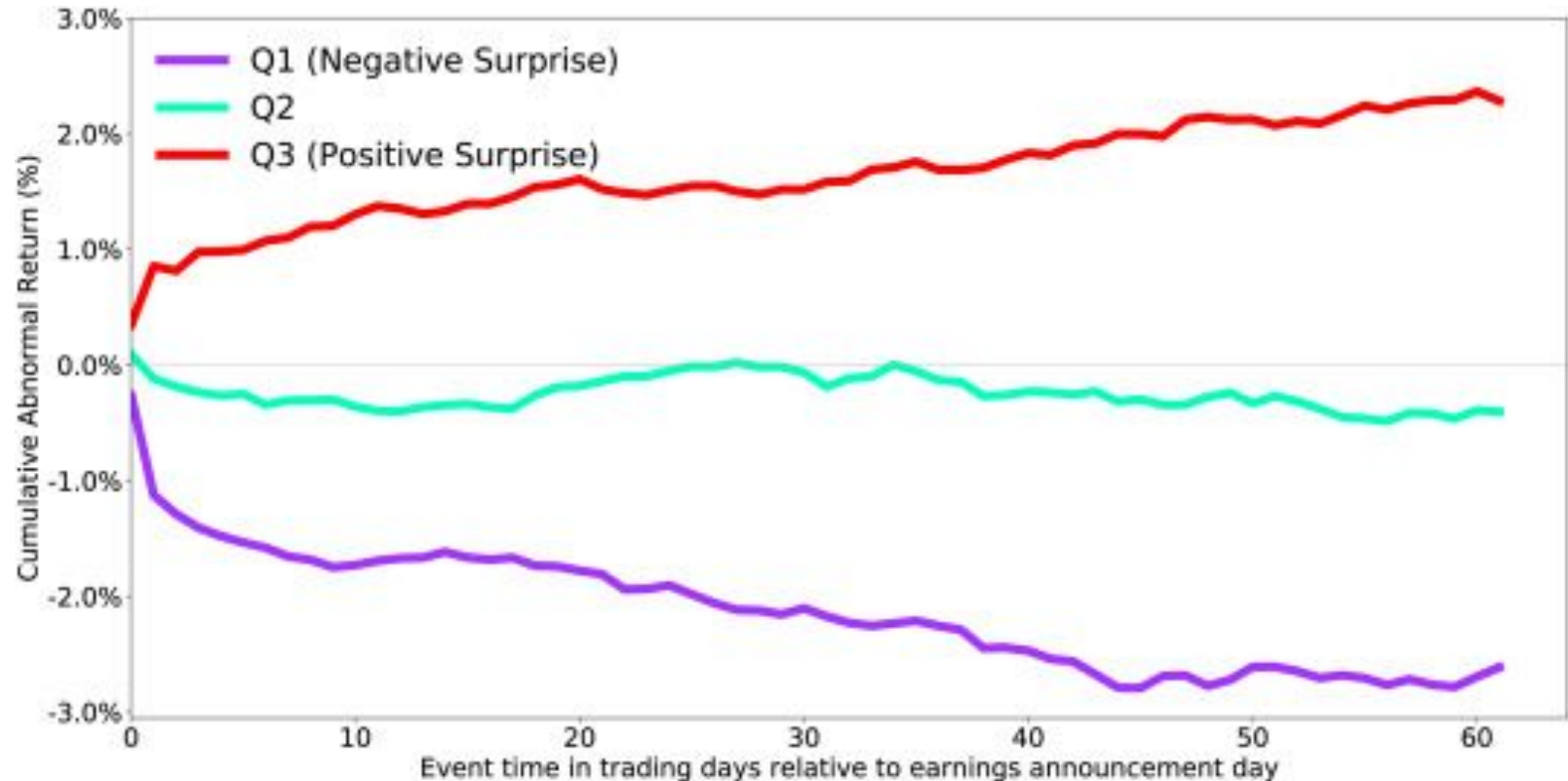
+0.54 (2.75%) ↑ past 6 months

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After hours 20.09 -0.10 (0.50%)



Technical Definition of Post-Earnings Announcement Drift



LULU Shows Downward Drift After Negative Jump

Market Summary > Lululemon Athletica Inc

196.45 USD

-216.35 (-52.41%) ↓ past 6 months

Closed: Aug 4, 7:53 PM EDT • Disclaimer

After hours 196.25 -0.20 (0.099%)

